

PFSENSE framework

1) General Information

- **Assessment Date:**23/4/2026
- **Analyst Name:** Mustafa Jbili
- **Scope of Work:** Cybersecurity Analyst
- **Objective :** A comprehensive report detailing the installation, configuration, and testing of a pfSense virtual firewall enforcing a default-deny inbound security policy.

2)introduction :

- This lab report explains how to set up a pfSense virtual firewall. The main goal was to learn how firewalls work by building one in a safe, virtual environment. We used VMware Workstation to create a virtual machine and installed pfSense, which is free, open-source firewall software.
- The most important part of the lab was to enforce a strong security rule called "default-deny." This means the firewall blocks all incoming traffic from the internet by default. We then created very specific rules to allow only the traffic we need, like web browsing and DNS lookups. This approach is a core best practice in cybersecurity to protect networks from attacks.
- By completing this lab, we learned the steps to provision a virtual firewall, configure its network interfaces, design a secure ruleset, and test that the rules work correctly. The final setup acts as a protective barrier, allowing good traffic in and out while stopping anything suspicious.

2) Lab's Environment :

- **Machine:** VMWARE WORKSTATION,kali(host inside lan) , metasploite(to simulate wan) , pfsense

3) VM and pfSense Installation Steps :

The first step was to download the pfSense software image from the official Netgate website. We chose the current stable release to ensure reliability. This ISO file was then mounted as virtual installation media inside our VMware Workstation.

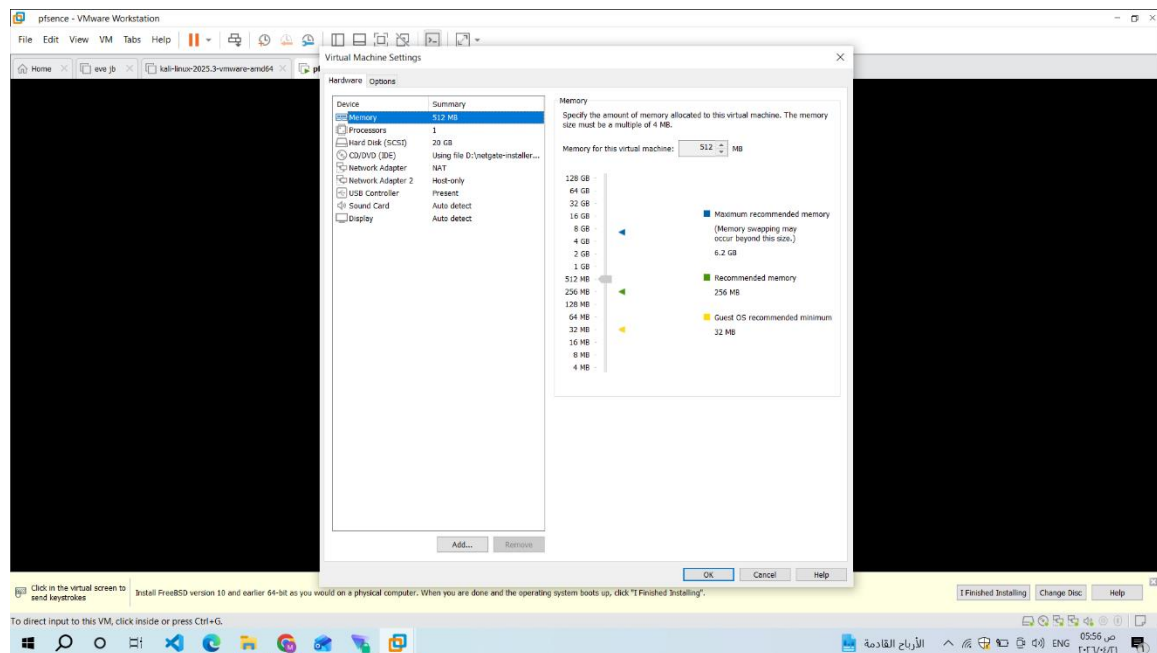
We created a new virtual machine with specific settings. The VM was given 512 MB of RAM and a 20 GB virtual hard disk to ensure smooth operation. The most critical pre-installation step was adding two virtual network adapters to the VM's settings.

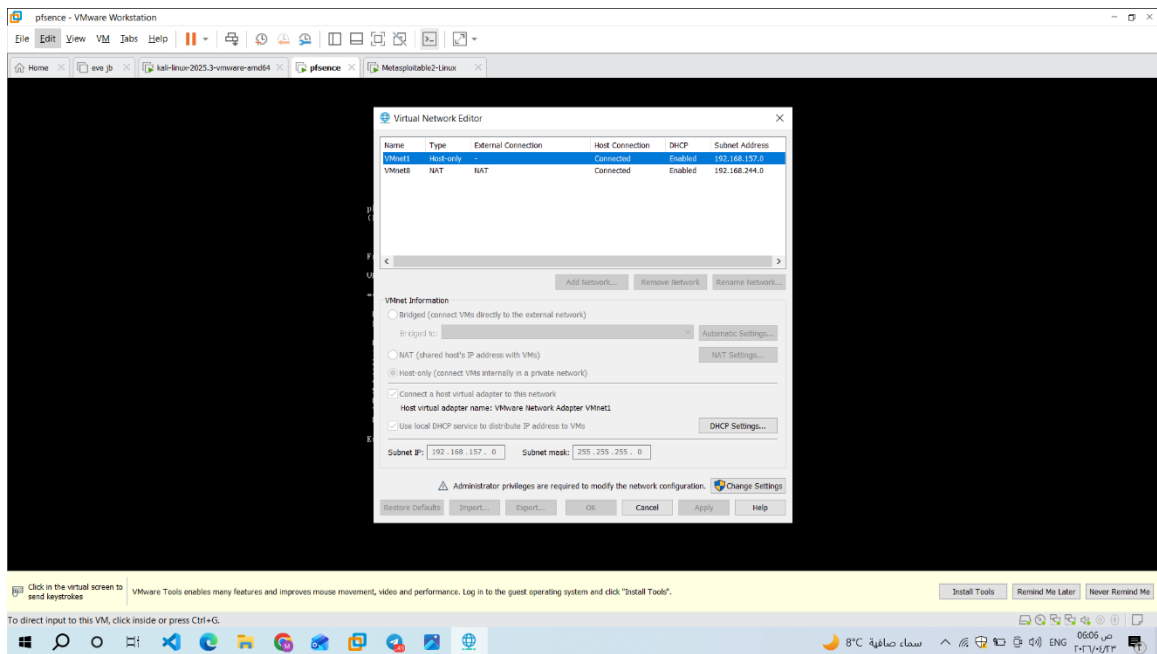
With the VM configured, we powered it on and booted from the pfSense ISO. The installer launched automatically. We selected the "Install" option and accepted all default choices, including installing the Community Edition (CE) and the "Current Stable Release." The installation process copied files to the virtual disk and completed within a few minutes.

After the install finished, we rebooted the VM and removed the installation ISO. The system then booted into the pfSense console, presenting a text-based menu ready for the initial network setup.

NETWORK ADAPTERS :

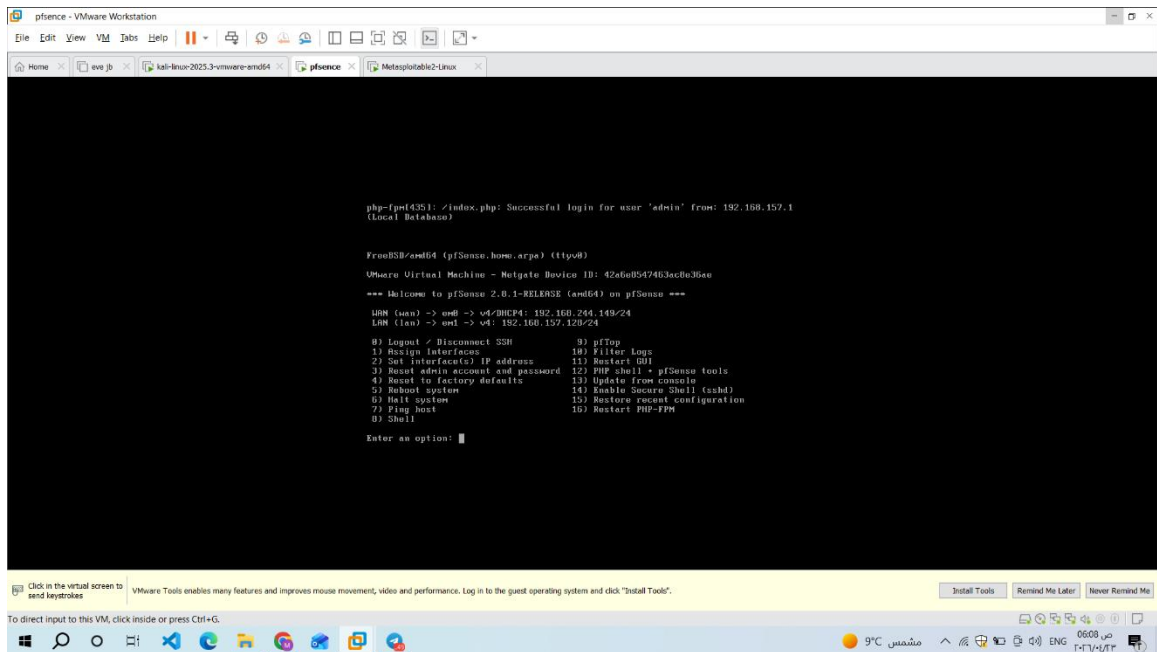
Configuring the network interfaces correctly is essential for the firewall to separate traffic. In the hypervisor, we pre-configured two network adapters. The first adapter was set to use NAT mode, simulating an external internet connection. This became our WAN (Wide Area Network) interface. The second adapter was set to Host-Only mode, creating an isolated internal network. This became our LAN (Local Area Network) interface.



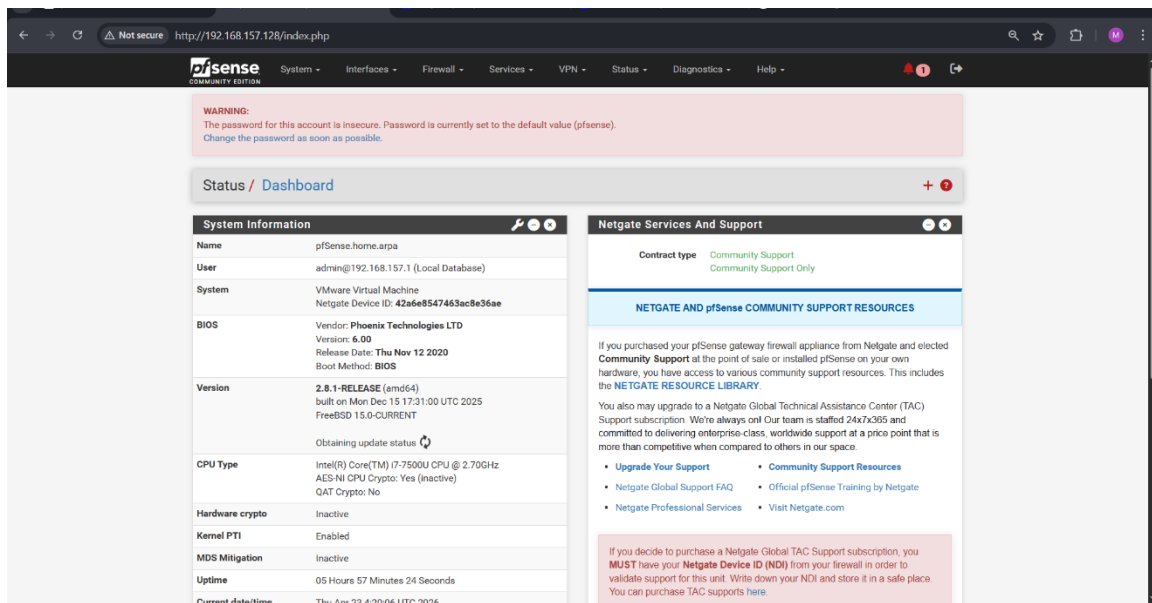


- First vmnet is for lan 192.168.157.0 and It was seized as host-only
- Second vmnet is for wan 192.168.244.0 and it was seized as Nat

4)Running pfsense :



- LAN and WAN network was configured to automatically obtain an AC address via the protocol DHCP



- Then successfully accessed to pfsense dashboard

5) Firewall Ruleset Design and Justification:

- The cornerstone of our security design is the default-deny inbound policy. This means any traffic trying to enter our network from the outside (WAN) is automatically blocked unless we have a specific rule to allow it. We removed any default "allow all" rules that pfSense might have created on the WAN interface to establish this strict baseline.
- On top of this deny-everything foundation, we built a minimal "allowlist." We only created rules for services that are absolutely necessary for the internal network to function. This includes outbound DNS for looking up websites, outbound HTTP and HTTPS for web browsing, and outbound NTP to keep system clocks accurate.
- We were very careful with ICMP (used by the 'ping' command). We allowed it only within the safe LAN for network troubleshooting but blocked all incoming ICMP from the WAN. This prevents attackers from using ping to discover our network. If remote management via SSH was needed, the rule would be restricted to a single trusted administrator's IP address, not open to the world.
- Every rule was made as specific as possible. We avoided using "any" for source or destination where we could. The rules are processed from top to bottom, so we placed our specific allow rules first. A final "block all" rule with logging enabled sits at the very bottom to catch and record any traffic that wasn't explicitly permitted.

Rule Table Summary :

Firewall Ruleset Summary

Inter face	Prot ocol	Source	Destinat ion	Port	Act ion	Justification
LAN	ICM P	LAN Net	LAN Net	Any	All ow	Permits internal network troubleshooting using ping.
LAN	TCP /UD P	LAN Net	Any	53 (DNS)	All ow	Allows internal clients to resolve domain names on the internet.
LAN	TCP	LAN Net	Any	80, 443 (HTTP/ S)	All ow	Permits outbound web browsing for internal users.
LAN	UDP	LAN Net	Trusted NTP Server	123 (NTP)	All ow	Allows internal systems to synchronize time with a reliable external source.
WA N	TCP	Trusted Admin IP	This Firewall	22 (SSH)	All ow	Permits secure remote management only from a specific trusted location.
WA N	Any	Any	Any	Any	Blo ck	Default-deny catch-all rule. Logs all blocked inbound traffic for review.

Wan :

pfSense
community edition

System • Interfaces • Firewall • Services • VPN • Status • Diagnostics • Help •

[+]

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfSense).
Change the password as soon as possible.

Firewall / Rules / WAN

🔍 📄 🚫

The firewall rule configuration has been changed.
The changes must be applied for them to take effect.

✓ Apply Changes

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	0/0 B	*	Reserved Not assigned by IANA	*	*	*	*	*	*	Block bogon networks	⚙️
☒	0/229 B	IPv4 *	*	*	*	*	none			rule to inbound traffic on the WAN interface default	📄 ✎️ 🔄 🚫
☒	0/0 B	IPv6 *	*	*	*	*	none			rule to inbound traffic on the WAN interface default	📄 ✎️ 🔄 🚫

➕ Add ➡️ Add 🔴 Deny 🔵 Triggers 📄 Copy 🗑️ Delete ➕ Secondary

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Lan :

Not secure http://192.168.157.128/firewall_rules.php?rl=lan

🔍 📄 🚫

WARNING:
The password for this account is insecure. Password is currently set to the default value (pfSense).
Change the password as soon as possible.

Firewall / Rules / LAN

🔍 📄 🚫

The changes have been applied successfully. The firewall rules are now reloading in the background.
Monitor the filter reload progress.

🔄

Floating WAN LAN

Rules (Drag to Change Order)

☐	States	Protocol	Source	Port	Destination	Port	Gateway	Queue	Schedule	Description	Actions
☒	5/744 KIB	*	*	*	LAN Address	80	*	*		Anti-Lockout Rule	⚙️
☒	0/229 B	IPv4 TCP/UDP	*	*	*	*	*	none		Allow DNS resolution	📄 ✎️ 🔄 🚫
☒	0/0 B	IPv4+6 ICMP echo(s)	LAN address	*	LAN address	*	*	none		Allow internal network ping for diagnostics	📄 ✎️ 🔄 🚫
☒	0/0 B	IPv4 TCP	LAN address	80 (HTTP)	*	*	*	none		Allow controlled web browsing traffic	📄 ✎️ 🔄 🚫
☒	0/0 B	IPv4 TCP	LAN address	443 (HTTPS)	*	*	*	none		Allow controlled web browsing traffic	📄 ✎️ 🔄 🚫
☒	0/0 B	IPv4 TCP	192.168.25.2	22 (SSH)	This Firewall (self)	*	*	none		Allow secure remote management from trusted IP only	📄 ✎️ 🔄 🚫
☒	0/0 B	IPv4 UDP	LAN address	123 (NTP)	*	*	*	none		Allow time synchronization with trusted NTP servers	📄 ✎️ 🔄 🚫

➕ Add ➡️ Add 🔴 Deny 🔵 Triggers 📄 Copy 🗑️ Delete ➕ Secondary

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6) Testing and Verification Results:

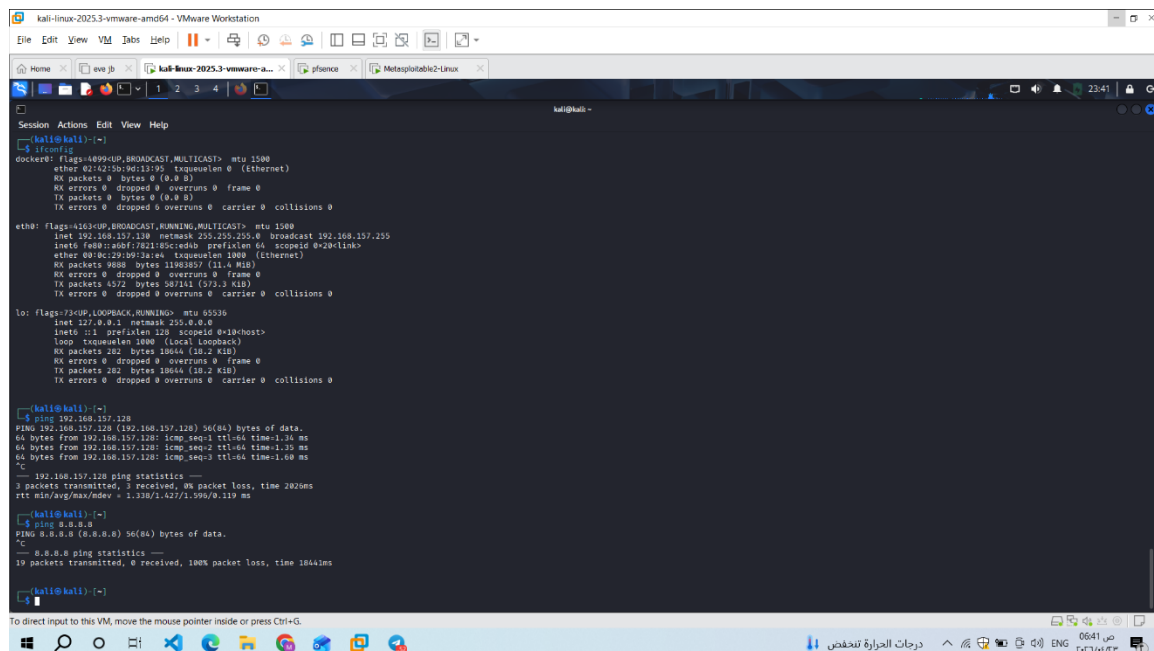
Testing proved our firewall rules worked as designed. From a host inside the LAN, we successfully performed a DNS lookup to google.com, confirming the outbound DNS rule was functional. We then opened a web browser and connected to an external HTTPS site, verifying the HTTP/HTTPS allow rules.

We also tested internal ICMP by pinging the firewall's LAN IP (192.168.1.1) from another machine on the same network. This succeeded, showing the LAN ICMP rule was correct. The firewall logs recorded these successful outbound connections as "pass" events.

The most critical test was for the default-deny rule. We attempted to ping the firewall's external WAN IP address from a simulated external attacker. The ping requests timed out and received no reply, proving inbound ICMP was successfully blocked. We also tried to connect to a non-existent service on the WAN IP.

The firewall log immediately showed a new entry for this attempt with a "block" action. The log entry included the source IP, destination IP, and port number, providing clear evidence that the catch-all deny rule was active and logging unauthorized access attempts.

1- internal icmp :



```
kali@kali:~$ ifconfig
docker0: flags=4096<UP,BROADCAST,MULTICAST> mtu 1500
    ether 02:42:5b:9d:13:95 txqueuelen 0 (Ethernet)
    RX packets 0 bytes 0 (0.0 B)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 0 bytes 0 (0.0 B)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

eth0: flags=4163<UP,BROADCAST,RUNNING,MULTICAST> mtu 1500
    inet 192.168.157.119 netmask 255.255.255.0 broadcast 192.168.157.255
    inet6 fe80::aabb:7821:85c:ed4b prefixlen 64 scopeid 0<24-link>
    ether 08:00:27:09:2a:0e txqueuelen 1000 (Ethernet)
    RX packets 9698 bytes 11983857 (11.4 MiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 4572 bytes 387141 (373.3 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
    inet 127.0.0.1 netmask 255.0.0.0
    inet6 ::1 prefixlen 128 scopeid 0<1<host>
    loop txqueuelen 1000 (Local Loopback)
    RX packets 282 bytes 18644 (18.2 KiB)
    RX errors 0 dropped 0 overruns 0 frame 0
    TX packets 282 bytes 18644 (18.2 KiB)
    TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

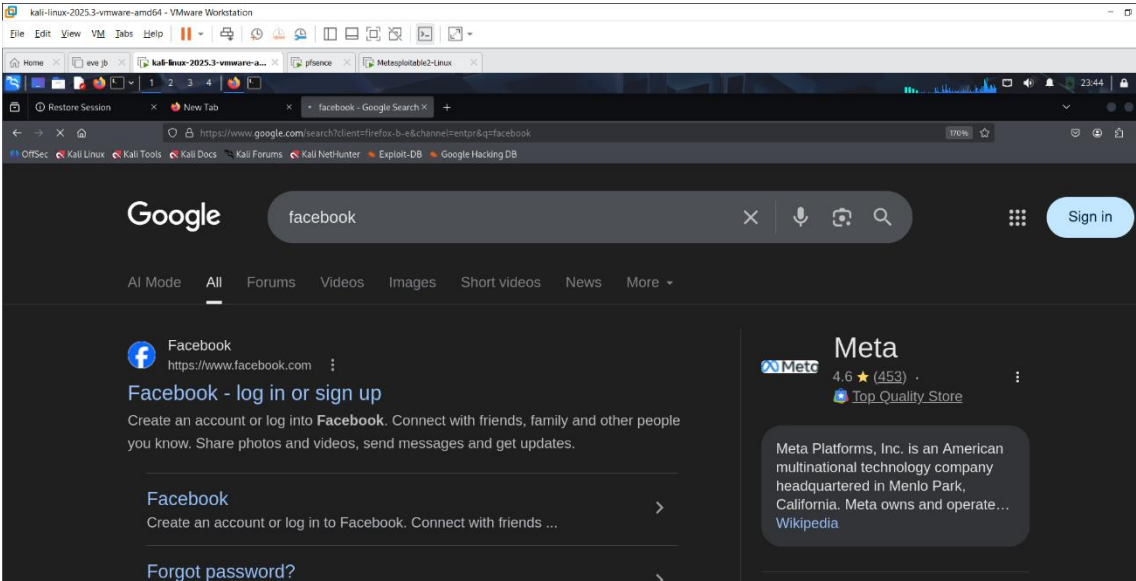
kali@kali:~$ ping 192.168.157.128
PING 192.168.157.128 (192.168.157.128) 56(84) bytes of data:
64 bytes from 192.168.157.128: icmp_seq=1 ttl=64 time=1.34 ms
64 bytes from 192.168.157.128: icmp_seq=2 ttl=64 time=1.25 ms
64 bytes from 192.168.157.128: icmp_seq=3 ttl=64 time=1.08 ms
^C
--- 192.168.157.128 ping statistics ---
3 packets transmitted, 3 received, 0% packet loss, time 2W2ms
rtt min/avg/max/mdev = 1.138/1.427/1.990/0.119 ms

kali@kali:~$ ping 8.8.8.8
PING 8.8.8.8 (8.8.8.8) 56(84) bytes of data:
^C
--- 8.8.8.8 ping statistics ---
19 packets transmitted, 0 received, 100% packet loss, time 1844ms
```

✖	Apr 23 03:39:34	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:33	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:32	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:31	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:30	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:29	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:28	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:27	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✖	Apr 23 03:39:26	LAN	Default deny rule IPv4 (1000000103)	🔒 192.168.157.130	🔒 8.8.8.8	ICMP
✔	Apr 23 03:39:15	LAN	👤 Allow internal network ping for diagnostics (1776743417)	🔒 192.168.157.130	🔒 192.168.157.128	ICMP
✔	Apr 23 03:34:09	LAN	👤 Allow time synchronization with trusted NTP servers (1776743911)	🔒 192.168.157.130:44468	🔒 95.216.192.15:123	UDP

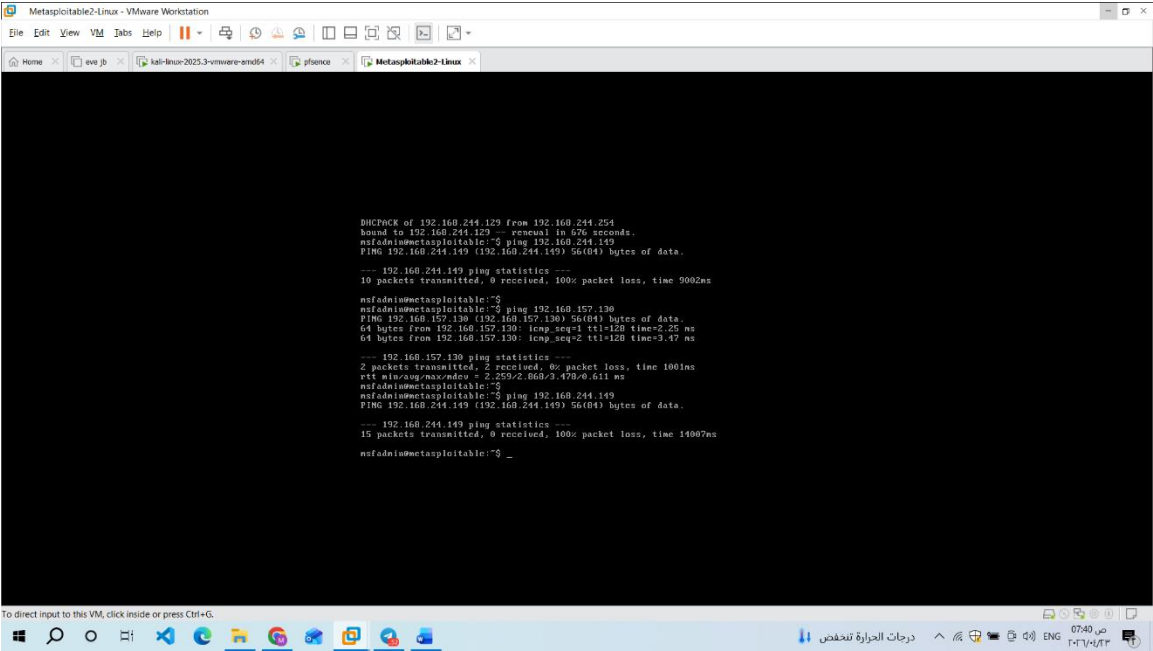
[illegible]

3-http/https :



✓	Apr 23 03:44:46	LAN	Allow controlled web browsing traffic (1776743737)	i 192.168.157.130:55394	i 142.250.181.168:443	TOP-S
✓	Apr 23 03:44:46	LAN	Allow controlled web browsing traffic (1776743737)	i 192.168.157.130:55380	i 142.250.181.168:443	TOP-S
✓	Apr 23 03:44:46	LAN	Allow DNS resolution (1776742730)	i 192.168.157.130:42024	i 192.168.157.128:53	UDP

4-wan testing :



Last 500 Firewall Log Entries. (Maximum 500)						
Action	Time	Interface	Rule	Source	Destination	Protocol
✗	Apr 23 04:40:18	WAN	rule to inbound traffic on the WAN interface default (1776742337)	192.168.244.129	192.168.244.149	ICMP
✗	Apr 23 04:40:17	WAN	rule to inbound traffic on the WAN interface default (1776742337)	192.168.244.129	192.168.244.149	ICMP
✗	Apr 23 04:40:16	WAN	rule to inbound traffic on the WAN interface default (1776742337)	192.168.244.129	192.168.244.149	ICMP
✗	Apr 23 04:40:15	WAN	rule to inbound traffic on the WAN interface default (1776742337)	192.168.244.129	192.168.244.149	ICMP
✗	Apr 23 04:40:14	WAN	rule to inbound traffic on the WAN interface default (1776742337)	192.168.244.129	192.168.244.149	ICMP